

Rare-earth ion doped on-chip amplifiers in the Al_2O_3 platform

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Abstract *Reactive sputtering of rare-earth-ion-doped Al_2O_3 enables scalable on-chip optical amplification across multiple wavelength bands. Monolithic integration with passive photonic platforms, combined with low nonlinearities and high-power handling, supports applications in telecommunications, datacom, sensing, and quantum photonics. Recent progress using erbium, thulium, and neodymium is reviewed. ©2026 The Author(s)*

Introduction

The increasing complexity and scale of photonic integrated circuits (PICs) for telecommunications, datacom, quantum technologies, sensing, and LiDAR has created a growing demand for compact and energy efficient optical amplifiers. On-chip amplification is essential for compensating propagation losses, extending transmission reach, enabling advanced laser integration, and supporting advanced photonic functionalities.

Nowadays semiconductor optical amplifiers (SOAs) are among the most widely used integrated amplifiers due to their compact footprint and compatibility with semiconductor fabrication technologies. However, SOAs suffer from several intrinsic limitations, including high noise figures, nonlinear distortions, polarization sensitivity, and strong temperature dependence. These characteristics become increasingly problematic in applications such as coherent communications and quantum photonics, where low noise and stable operation are critical.

Rare-earth-ion-based gain media offer an attractive alternative. Rare-earth ions (RE^{3+}) have enabled high-performance optical amplification in fiber systems most notably through erbium-doped fiber amplifiers (EDFAs), which form the backbone of long-haul optical telecommunications [1]-[3]. Compared with semiconductor optical amplifiers, rare-earth-ion-based amplifiers exhibit lower noise, reduced nonlinear distortions, and improved thermal stability.

However, conventional fiber amplifiers are inherently bulky and difficult to scale toward dense photonic integration. Rare-earth-ion-doped waveguide amplifiers emerge as a promising technology combining the excellent spectroscopic properties of RE^{3+} ions with the compactness and scalability of integrated photonics [4]-[5].

Several material platforms have recently demonstrated integrated rare-earth ion based amplification. Erbium-doped thin-film lithium

niobate waveguides have shown significant optical gain [6]. Nevertheless, this material typically relies on heterogeneous integration with passive photonic platforms, limiting scalability and fabrication simplicity [7]. Alternatively, amplification in Si_3N_4 has been achieved using high-energy erbium ion implantation [9]-[10] at the expense of significantly increased fabrication complexity and with limitations such as elevated polarization-dependent gain [11]. In both platforms, nonlinear optical effects also impose restrictions on achievable operating powers.

Aluminium oxide (Al_2O_3) addresses many of these challenges by combining excellent optical properties with scalable fabrication. Rare-earth ion doped Al_2O_3 films are deposited directly doped using wafer-scale reactive co-sputtering [12], enabling uniform incorporation of different rare-earth ions throughout the film. The sputtering process naturally supports monolithic integration with passive waveguide platforms. In addition, Al_2O_3 exhibits a very low nonlinear refractive index [13], low propagation losses, and excellent thermal and power handling capabilities, particularly in its polycrystalline phase [12].

The flexibility of the $\text{Al}_2\text{O}_3\text{:RE}^{3+}$ platform enables optical amplification across multiple wavelength bands simply by changing the dopant ion. This invited paper reviews recent progress in erbium-, thulium-, and neodymium-doped Al_2O_3 waveguide amplifiers and discusses future directions toward even broader wavelength coverage.

Erbium doped Al_2O_3 waveguide amplifiers

The development of $\text{Al}_2\text{O}_3\text{:Er}^{3+}$ waveguide amplifiers was initially motivated by the need for compact amplifiers operating across the telecom C-band. Through continued optimization of the deposition and fabrication processes, photonic integrated circuits with high internal gain (i.e., up to 3.5 dB/cm) [14] and over 24 dB of external net gain with a gain bandwidth exceeding 90 nm [15]-[16] have been demonstrated. The amplifier

maintains stable operation up to temperatures of 95 °C, highlighting a major advantage over SOAs, which generally require active thermal stabilization. By careful design of the amplifier waveguide cross-section and edge couplers, devices with negligible polarization dependent gain (PDG) can be obtained.

The combination of broad bandwidth, high gain, thermal robustness, and compatibility with integrated photonics makes $\text{Al}_2\text{O}_3:\text{Er}^{3+}$ highly attractive for coherent communication systems, future optical interconnects and quantum technology.

High power thulium amplifiers and lasers

The versatility of reactive sputtering is further demonstrated using thulium (Tm^{3+}) as the active dopant, which provides broadband emission in the 1.8–2.0 μm wavelength range, highly relevant for sensing, spectroscopy, and LiDAR.

$\text{Al}_2\text{O}_3:\text{Tm}^{3+}$ was integrated with Si_3N_4 to form a hybrid waveguide that permits realizing large mode area (LMA) amplifiers, which have shown optical output powers exceeding 1 W and short pulse amplification (i.e., 116 fs pulses) to peak powers of 800 W [17]–[20].

Neodymium doped waveguide amplifiers

Neodymium ions are attractive due to their emission bands around 880 nm, 1064 nm, and 1310 nm, opening opportunities for biomedical imaging, spectroscopy, quantum technologies, and O-band datacom applications.

Promising performance has already been demonstrated in amorphous $\text{Al}_2\text{O}_3:\text{Nd}^{3+}$, with reported gains of 1.57 dB/cm at 880 nm, 6.30 dB/cm at 1064 nm, and 1.93 dB/cm at 1310 nm [21]. Amplification in the O-band is particularly relevant for next-generation short-reach optical interconnects.

Current research focuses on polycrystalline $\text{Al}_2\text{O}_3:\text{Nd}^{3+}$, which is expected to provide enhanced absorption and emission cross-sections while suppressing ion clustering effects that limit gain in amorphous hosts. Furthermore, the high thermal budget of polycrystalline Al_2O_3 enables the use of advanced high-temperature cladding deposition processes, improving the performance of edge couplers and overall device efficiency.

A first demonstration of polycrystalline $\text{Al}_2\text{O}_3:\text{Nd}^{3+}$ waveguide amplifiers operating at 1064 nm achieved fiber-to-fiber gain values up to 11.9 dB in compact 5.5 cm long devices [22]. These results illustrate the strong potential of Nd-doped Al_2O_3 for compact high-performance amplifiers in emerging photonic systems.

Conclusions

Rare-earth-ion-doped Al_2O_3 waveguide

amplifiers fabricated by reactive sputtering provide a scalable and high-performance solution for integrated optical amplification. The platform combines low loss, low nonlinearities, excellent thermal stability, high power handling, and compatibility with monolithic photonic integration.

Recent demonstrations using erbium, thulium, and neodymium highlight the broad applicability of the platform across telecommunications, datacom, sensing, LiDAR, and quantum technologies. These results position $\text{Al}_2\text{O}_3:\text{RE}^{3+}$ as a highly promising candidate for next-generation photonic integrated circuits.

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